Example Report: Timing a Pendulum

Report Viewer example

1 Purpose

This short document exists so you can try the report viewer. Select a sentence and add a comment, press *Read aloud* and tap any word to hear the text from there, or ask a Claude Code session to answer your comments.

2 Method

A small mass hangs from a light string of length L. For small swings the period is

$$T = 2\pi\sqrt{\frac{L}{g}}, \tag{1}$$

where g is the acceleration due to gravity. The read-aloud skips display equations like this one and carries on with the next sentence. Timing ten swings with a stopwatch and dividing by ten reduces the effect of reaction time on each measurement.

3 What to try

Change the length of the string, time the swings again, and plot the period against the square root of the length. The points should fall on a straight line through the origin, and its slope gives an estimate of g. Write down the uncertainty of every timing, because the scatter between repeated measurements shows how far the estimate can be trusted.